

▷ CRATON REVEALED

In the Drakensberg, South Africa, the Blyde River has cut down to the ancient rocks of the Kaapvaal Craton.



THE SHAPING OF AFRICA

Africa is formed around five stable sections of Earth's crust called cratons. They were the heart of the giant southern continent Gondwana during the age of the dinosaurs. About 65 million years ago, the land around these cratons broke away to create Africa.

Ancient heart

The five African cratons were formed over a billion years ago when magma welled up from Earth's interior and solidified. In places, they have since been altered by heat and pressure, and in others, they

have been covered by a veneer of sediments. The rocks they are made of are very tough—they have endured as other rocks came and went, and continents and oceans shifted around them. The cores of the cratons are ancient and date

BUILDING BLOCKS

Africa is built around five major cratons of ancient rock, with the most ancient of all being the Kaapvaal Craton to the south. The Arabian-Nubian Craton is being rifted apart.



back to one of the oldest periods in Earth's history, the Archean Era, over 2.5 billion years ago. The

oldest rocks are the granite, gneiss, and greenstone terrains of parts of South Africa, Zimbabwe, and Tanzania. The Kaapvaal Craton is noted for its distinctive komatiite rocks that formed when Earth was very hot in its earliest days. The area is also famous for its deposits of gold, nickel, and uranium.

Mountain fringes

For long periods in Earth's history, the African cratons were separate continents. A billion years ago, the craton to the west was attached to a landmass that would eventually hold the Amazon Basin within an ancient supercontinent, while the central and southernmost cratons were islands. Gradually, they were all glued together by mobile belts of rock and became part of the Pangaea supercontinent 250 million years ago. Meanwhile, the mobile belts were crunched between the cratons, folding up rocks to form mountain chains such as the Atlas and the Cape Fold Mountains, some of which survive in South Africa (others are in South America, Australia, and Antarctica).

KEY EVENTS

140 million years ago The landmass that will become Africa splits from what will become South America due to the opening of the Mid-Atlantic Rift, creating a new ocean—the Atlantic.

30 million years ago The Arabian Peninsula pushes away from Africa as magma forces through Earth's crust and cools, creating new land that is denser than the older, surrounding land. As a result of this higher density, it sinks—creating the Afar Depression.

1.8 million years ago Mount Kilimanjaro forms due to volcanic activity linked to the opening of the Great Rift Valley.

160 million years ago Madagascar (along with the Seychelles and India) splits from Africa, and then about 84 million years ago it splits again from the Seychelles and India—both events are due to rifting as the Indian Ocean opens up.

30 million years ago Many faults open between various plates in eastern Africa, and the land between three neighboring plates “drops” to form the Great Rift Valley.

2.5 million years ago North Africa becomes increasingly dry, eventually forming the Sahara Desert. Changes in the tilt of Earth's axis lead to brief periods of increased humidity and plant growth.

400,000 years ago Tilting of Earth's crust creates a basin. Water collects within it and Lake Victoria forms. During the ice age it drains away multiple times as the rate of rainfall rises and falls.